ChatGPT

AI4Science: A Comprehensive Landscape Review (2025)

Artificial Intelligence for Science (AI4Science) has rapidly grown into a broad interdisciplinary field. AI techniques are now **augmenting scientific discovery** by helping researchers generate hypotheses, design experiments, analyze complex datasets, and uncover insights beyond the reach of traditional methods [1]. Over the past decade, breakthroughs like self-supervised learning (training on vast unlabeled data) and geometric deep learning (exploiting structural knowledge in data) have enhanced accuracy and efficiency in scientific tasks [2]. Generative AI can even design novel molecules and proteins by learning from images, sequences, and multimodal data [2]. Despite these advances, central challenges remain: scientists and AI practitioners must identify when and how AI approaches fail, improve data quality and curation, and develop **foundational methods that can autonomously acquire scientific understanding** [3]. This review surveys the current AI4Science landscape across academia and industry, highlighting key groups, major companies, core research themes (with priority to computer science applications, then biology/chemistry, then other sciences), the role of datasets, and current gaps and future directions.

Key Research Groups and Institutions

Academic interest in AI4Science is **worldwide and highly interdisciplinary**. Universities and institutes have formed dedicated groups that unite computer scientists with experts in physics, biology, chemistry, and more. For example, the U.S. National Science Foundation launched new AI Research Institutes focusing on scientific domains (e.g. AI for physics, astronomy, materials) to spur multidisciplinary collaboration [4]. One such institute, the **AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI)**, links MIT, Harvard, and others to advance physics knowledge with AI and develop AI techniques informed by physical principles [4]. Similarly, Cornell University's AI for Science Institute organizes research around themes like materials discovery, physics, biological sciences, and sustainability, emphasizing cross-domain "synergies" (quantum physics, biophysics, climate, etc.) [5] [6]. These initiatives illustrate academia's recognition that tackling grand scientific challenges often requires integrating AI expertise with domain knowledge.

Collaborative projects are also bringing together multiple institutions. The **Polymathic AI** project, launched in 2023, is a cross-disciplinary effort involving the Flatiron Institute (Simons Foundation), University of Cambridge, Princeton, Lawrence Berkeley National Lab, and others [7]. Led by astrophysicist Shirley Ho, Polymathic AI is building a *foundation model* for science that learns from diverse fields (initially physics and astrophysics data, with plans for chemistry and genomics) [8] [9]. The idea is that a model pretrained on many scientific "languages" can find connections and patterns across disciplines that specialists might miss [10]. This reflects a broader trend in academia: **preparing large, pre-trained AI models for science** and "democratizing" them by open-sourcing them to the scientific community [11].

Many universities now host **AI+Science labs** or centers. For instance, Caltech and the University of Chicago have hosted conferences on AI + Science to share advances [12], and institutions like MIT, Stanford, and University of Washington have research groups applying AI to protein engineering, materials synthesis, and

neuroscience. In Europe, initiatives like the European Lab for Learning & Intelligent Systems (ELLIS) include programs on AI for Science, and the UK's Turing Institute and Cambridge's Milner Institute are exploring how AI changes scientific research practices. In sum, academic research groups around the globe are actively bridging computer science with traditional sciences, producing influential papers and training a new generation of researchers fluent in both AI and domain specifics.

- *Example Academic Efforts:* NSF AI Institutes (e.g. IAIFI for physics [4]), Cornell AI for Science (materials, bio, physics) [6], Flatiron Institute's Polymathic AI (cross-disciplinary foundation model) [7], and numerous university labs focusing on AI-driven drug discovery, materials design, climate modeling, etc.

Major Companies and Industrial Labs

Industry has heavily invested in AI4Science, with **big tech companies and specialized startups** tackling scientific problems using AI at scale. **Google DeepMind** (and Google Research) has led several headline-making projects. Notably, DeepMind's *AlphaFold* achieved a breakthrough in protein structure prediction using deep learning, essentially **solving the 50-year protein folding problem** by predicting 3D structures from sequences with atomic accuracy [13]. AlphaFold's Transformer-based AI system revolutionized biology and was recognized as a transformative milestone in science [13]. DeepMind has since open-sourced the AlphaFold database of predicted structures, and its success has spurred many follow-ups in bioinformatics and beyond. Alphabet/Google's AI teams also applied AI to pure mathematics and algorithms – e.g. *AlphaTensor* discovered improved algorithms for matrix multiplication via reinforcement learning (an achievement in theoretical computer science), and collaborations with mathematicians used deep networks to conjecture new theorems in knot theory and representation theory (published in *Nature*, 2021). In climate science, **Google DeepMind's GraphCast** model demonstrated that AI can outperform traditional supercomputer simulations for weather forecasting: GraphCast learned from 40 years of global weather data and can predict 10 days ahead in seconds, beating the accuracy of the world's best numerical weather models on 90% of key metrics [14] [15]. This is considered a "turning point" for forecasting, achieving comparable or better accuracy at a tiny fraction of the computation cost [14] [15]. Google has even prototyped an **"AI co-scientist"** system – a multi-agent AI (built on their Gemini model) that generates and evolves scientific hypotheses. In early biomedical tests, this AI co-scientist proposed viable drug repurposing candidates and new disease targets which were validated in lab experiments [16] [17]. These examples show Google's broad push: from biology and medicine to math and earth science, often leveraging large-scale models and vast compute.

Microsoft Research established a dedicated global *AI4Science* team in 2022 [18]. Led by Chris Bishop and colleagues (including renowned scientist Dr. Frank Noé), this team spans experts in machine learning, **quantum physics, computational chemistry, molecular biology, and fluid dynamics** [19]. Microsoft's AI4Science projects target fundamental science challenges: one project, **Graphormer**, is a deep learning toolkit for molecular modeling (based on graph neural networks). Graphormer won the Open Catalyst Challenge by accurately predicting catalyst reaction outcomes, leveraging a massive dataset of *0.66 million* simulated catalyst structures (144 million atomic frames from quantum chemistry calculations) [20] [21]. Another project in collaboration with Novartis is **Generative Chemistry**, using AI to design new drug molecules and accelerate medicinal chemistry [22]. Microsoft emphasizes the "*fifth paradigm*" of science – *AI-powered simulation emulation* – where neural networks learn from high-fidelity physics simulations (e.g. solving Schrödinger's equation for molecular quantum mechanics) to create fast surrogates that maintain accuracy [23] [24]. This approach can speed up tasks like drug binding affinity prediction or climate modeling

by orders of magnitude [24]. The Microsoft AI4Science team's breadth and partnerships (including with pharma) underscore industry's interest in applying AI both for scientific discovery and to practical R&D challenges.

Other major companies are active as well. **IBM Research** has long worked on AI for chemistry and materials (e.g. the IBM RXN platform uses AI to predict chemical reactions and aid synthesis planning). IBM's projects include using deep generative models to suggest new molecular structures and applying AI in quantum computing and physics simulations. **Meta (Facebook) AI** built large protein language models (like ESM-2) that can predict structures and functions; Meta's ESMFold predicted hundreds of millions of new protein structures from metagenomic sequences, complementing AlphaFold's work. Meta also explored **scientific LLMs** (though their Galactica science LLM highlighted the challenge of accuracy in AI-generated scientific text). **Amazon** has applied AI in its AWS cloud for genomics analysis and launched initiatives in climate forecasting and supply chain optimization (related to AI for science of complex systems), and is involved in healthcare AI. Beyond Big Tech, **specialized startups and labs** are prolific in AI4Science: for example, *BenevolentAI* and *Insilico Medicine* (drug discovery startups) use generative models and AI-driven lab tests to find new therapeutics [25]. *DeepMind's spin-off Isomorphic Labs* is focusing on AI for drug design. In materials and energy, companies like *Schrödinger* and *OpenAI's collaboration with Total* have used AI to search for new catalysts and battery materials. Even the automotive and aerospace industries (e.g. Tesla, Boeing) apply AI to materials and fluid dynamics problems for design optimization.

In summary, **industrial labs are pushing AI4Science at scale**, often achieving landmark results by combining massive datasets, computing power, and domain expertise. Tech companies also frequently partner with academic or government labs (for instance, Microsoft with Novartis, or Google with European Molecular Biology Lab for AlphaFold) to validate and deploy AI discoveries. This synergy is accelerating progress and moving AI4Science innovations from research to real-world impact (e.g. AlphaFold's predictions now guide experimental biologists worldwide).

- *Example Industry Efforts:* Alphabet/Google DeepMind (AlphaFold for protein folding [13]; GraphCast for weather [14]), Microsoft AI4Science (Graphormer for catalysts [20]; generative drug design), Meta AI (protein language models), IBM Research (AI-driven chemistry), and AI-driven startups in biotech and materials [25].

Core Research Themes and Recent Advances

Under the AI4Science umbrella, research spans many scientific disciplines. Below we outline major themes and advances, prioritizing **computer science-driven applications** first, then **biology/chemistry**, followed by **other physical sciences**:

AI in Computing, Mathematics, and Simulation Science

One important theme is using AI to tackle **problems within computer science and mathematics themselves**, and to supercharge scientific computing. A prominent example is *AI-designed algorithms*: DeepMind's **AlphaTensor** used reinforcement learning to discover a new faster algorithm for matrix multiplication (a fundamental computing task), outperforming centuries-old human-derived algorithms. This was a breakthrough in algorithmic mathematics, hinting that AI can help invent more efficient computational methods. In a related vein, **code generation models** like DeepMind's AlphaCode have solved complex competitive programming problems, suggesting AI could aid in writing software for

scientific simulations or data analysis. These developments apply AI to what are traditionally computer science challenges, directly advancing the tools available for all sciences.

Another crucial area is **AI as a surrogate for simulations**. Scientists often rely on expensive numerical simulations in fields like climate modeling, fluid dynamics, cosmology, or quantum mechanics. AI models can learn from the input-output of these simulations and then emulate them much faster. This is the basis of the "fifth paradigm" of scientific discovery, which focuses on training deep learning models on data generated by fundamental physics equations [23]. The *GraphCast* weather model is a striking success of this approach – after training on decades of simulation-based weather reanalysis data, the GraphCast GNN predicts global weather in <1 minute with higher accuracy than a 6-hour traditional simulation [14] [15]. Similarly, in plasma physics, AI controllers have learned to manipulate fusion reactor plasmas in real-time (a DeepMind collaboration on tokamak control). In quantum chemistry, neural networks (like Ferminet or PauliNet) trained on the Schrödinger equation can predict molecular energies at near quantum-accurate levels much faster than classical computational chemistry. By treating *simulation outputs as perfectly labeled data*, these AI models **bridge computer science and natural science** – speeding up computation by orders of magnitude while maintaining fidelity [24]. This enables researchers to explore far larger design spaces (e.g. screening millions of candidate materials or drug molecules) than previously possible.

Moreover, AI is being integrated with **theorem proving and symbolic mathematics**. Automated theorem provers enhanced with neural guidance (e.g. DeepMind's work on the Lean theorem prover) have started to solve and even generate new conjectures. In 2022, an AI system helped mathematicians conjecture a new relationship in representation theory by sifting through big datasets of mathematical objects – demonstrating that pattern-finding AI can assist human "deep thinking" in math. Though nascent, this partnership of **deep learning + deep reasoning** hints at AI contributing to formal scientific knowledge (like discovering new mathematical truths or laws).

Finally, **computer science methods like knowledge representation, search, and optimization** are core to many AI4Science advances. Graph neural networks, for example, provide a natural way to represent molecules, materials, or networks in biology; their development (a computer science task) directly benefits chemistry and bio applications. Reinforcement learning, a CS subfield, has been used not just in games but in optimizing lab experiment sequences or designing DNA sequences with desired functions. Advances in **causal inference and probabilistic programming** (AI fields overlapping with stats) are being applied to understand cause-effect in complex scientific data (like disentangling factors in climate data or epidemiology). In summary, the computer science side of AI4Science delivers the *algorithms, models, and computational innovations* that power applications in all other scientific domains.

AI in Biology and Medicine

Biology and medicine have been hugely influenced by AI, with numerous breakthroughs in the past few years. The flagship achievement is **protein structure prediction**. DeepMind's *AlphaFold* and related models (RoseTTAFold, Meta's ESMFold) now predict 3D structures of proteins from amino-acid sequences with atomic-level accuracy [13]. This capability, once thought to be decades away, was validated in the CASP competition and is transforming structural biology. Millions of protein structures have been predicted and released, enabling biologists to understand protein functions and interactions in ways that used to require laborious crystallography experiments [13]. Following AlphaFold, researchers are tackling the next challenges: predicting protein *complexes* (how multiple proteins assemble), *dynamics* (how structures move), and designing new proteins from scratch. Here, generative AI is key – models like ProteinDiffusion and

ProGen can generate novel protein sequences that fold into specified shapes or perform desired tasks (for example, enzymes with new functions). This overlaps with **drug discovery**, where AI is used to design or screen new therapeutics. Deep learning models can predict whether a molecule will bind to a target protein (virtual screening), suggest chemical modifications to improve a drug candidate, or even generate novel molecular structures with certain properties (using methods like variational autoencoders or diffusion models) [25]. Several drug candidates designed with the help of AI are now in preclinical trials [22] [26], and big pharma companies have partnered with AI startups to accelerate medicinal chemistry.

AI is also advancing **genomics and medicine**. In genomics, deep learning models (like DeepVariant, AlphaMissense) can identify mutations in DNA and predict their effects, helping pinpoint disease-causing genetic variations. Large language models trained on DNA or protein sequences (sometimes called "Genomic Transformers") capture patterns of evolution and function, aiding in annotating the genome's "dark matter." In healthcare, AI systems are improving medical imaging analysis (for instance, diagnosing cancers from scans or retinal diseases from eye images). They also assist in analyzing electronic health records to find patterns across patient populations and predict outcomes. Notably, **multi-modal bioscience AI** is rising: models that combine genetic data, medical imaging, and clinical notes to provide a more holistic prediction (for example, predicting disease progression by integrating different data sources).

Another exciting area is **single-cell biology and bioinformatics**. Techniques like single-cell RNA sequencing produce enormous, complex datasets of gene expression; AI clustering and embedding methods help find structure in this data (identifying cell types, developmental trajectories, etc.). For instance, specialized AI methods have been developed for *genomic sequence analysis* or *protein design*, tailored to these domains [27]. The scientific community also uses AI for tasks like predicting protein folding pathways, modelling ecosystems (in computational ecology), and analyzing brain imaging in neuroscience (e.g. extracting features from MRI scans to diagnose neurological conditions).

However, the integration of AI in biology/medicine also highlights important challenges. Ensuring *explainability* and *trustworthiness* of models is crucial – clinicians and biologists need to understand an AI's predictions. There is active research on **interpretable AI for biology**, such as methods that highlight which gene or protein features most influence a model's decision. Additionally, issues of data bias and quality are prominent: medical and biological data can be noisy or unbalanced, so researchers use techniques like data augmentation (e.g. generating synthetic cell images) and rigorous validation to improve reliability [3]. Despite these challenges, the trajectory is clear: AI is becoming an indispensable tool in life sciences, from early research through to clinical applications, dramatically speeding up discovery (as seen with AlphaFold) and opening new avenues (like AI-designed enzymes, personalized medicine predictions, and more).

AI in Chemistry and Materials Science

In chemistry and materials science, AI is empowering scientists to explore chemical space and design new substances with unprecedented scope. **Molecular property prediction** was one of the early successes – given a chemical structure, ML models can predict properties like solubility, reactivity, or toxicity much faster than lab tests. This helps filter candidates in drug and material design. Recently, the focus has shifted to **generative chemistry**: using AI to propose entirely new molecules or materials that meet certain design criteria. Models such as graph-generative networks or transformers can output novel chemical formulas optimized for, say, high battery capacity or specific drug-target interactions [22]. These suggestions then guide chemists on what to synthesize next. Generative AI, combined with active learning, creates an

iterative loop where the model suggests a molecule, experiments test it, data is fed back to retrain the model, and so on – a closed-loop optimization of chemical space.

AI is also deeply integrated into **materials science** research. Material design problems (like finding a catalyst for CO_2 capture or a semiconductor with specific properties) often involve vast compositional possibilities. *Materials informatics* applies ML to predict material properties from composition or to detect patterns in crystallographic data [28]. For example, neural networks have been trained on databases of known materials (and simulation results from DFT calculations) to predict band gaps, conductivity, or strength of new materials [28]. A 2021 study by Guo *et al.* showed AI and machine learning methods accelerating the design of mechanical metamaterials, demonstrating how algorithms can suggest structures with desired mechanical properties [28]. Additionally, AI aids in analyzing spectroscopy and microscopy data – tasks like identifying molecules from their mass spectra or classifying crystal structures from microscopy images are now done with ML classifiers.

Another revolutionary development is the **autonomous lab or "robot scientist."** Researchers have built robot systems that use AI to plan and perform experiments with minimal human input. A striking example comes from the University of Liverpool: a 1.75 m tall mobile robot chemist that can navigate a laboratory, handle equipment, and run experiments 24/7.

A mobile robot scientist in a chemistry lab. Such autonomous robots can perform experiments around the clock and have even discovered new catalysts [29]. This robot uses AI to decide which experiments to try – in a proof-of-concept, it conducted 688 experiments over 8 days (over 170 hours of active work) and **independently discovered a new catalyst** that is six times more active than previous ones [29] [30]. Unlike traditional lab robots that follow pre-programmed steps, this system "thinks" about what to do next (within a predefined goal) and adapts, embodying a physical form of AI-driven discovery. Similar autonomous research systems have been reported: e.g. AI-driven synthesis machines that optimize conditions to grow high-quality crystals or find the best recipe for a photovoltaic material [31]. These **self-driving labs** accelerate research by freeing human scientists from tedious trial-and-error and by searching vast experimental spaces more efficiently than a human could.

In chemistry, AI is also tackling **reaction prediction and retrosynthesis** – essentially, teaching an AI chemist to plan how to make a molecule. With large datasets of known reactions, models can suggest reaction steps to assemble a target compound (like a GPS for chemical synthesis). This has become a valuable tool for chemists in drug development and manufacturing. Companies and academics have deployed AI assistants that propose synthetic routes and even control automated flow reactors to execute them.

Materials science also benefits from **physics-informed AI**. Researchers often incorporate known physical laws or symmetries into models (for example, enforcing that a model's output for energy obeys conservation laws) to improve reliability. A notable thread is **AI for quantum materials and physics** – using ML to detect phase transitions in complex simulations or to steer quantum experiments. Even **battery science and chemistry for sustainability** are seeing AI contributions, such as predicting electrolyte behavior or discovering catalysts for green chemistry. All these efforts contribute to faster development of new materials and chemicals, which is crucial for innovations in energy, electronics, and pharmaceuticals.

AI in Physical Sciences, Engineering, and Other Fields

Beyond bio and chem, AI methods are permeating **physics, earth science, astronomy, and engineering**. We already discussed climate/weather modeling via GraphCast; more broadly, **climate science** has embraced AI for tasks like extreme weather prediction, climate downscaling (translating global climate model output to local effects), and analyzing climate sensor data [32]. For example, AI models can identify patterns in satellite imagery to monitor deforestation or ice-sheet changes, and climate scientists use neural nets to emulate cloud processes that are too small-scale for global simulations [33] [34]. In **astrophysics**, AI helps manage the deluge of data from telescopes: machine learning classifiers sift through petabytes of sky survey images to find rare events (like supernovae or gravitational lenses). Shirley Ho's group, for instance, uses deep learning to analyze cosmological simulations and observational data together, finding subtle signals of new physics. The gravitational wave observatories (LIGO/Virgo) employ ML to detect potential wave signals amidst noise. In **particle physics**, large experiments like the CERN LHC leverage AI to quickly filter interesting collision events and to reconstruct particle tracks; and physicists are exploring if generative models can simulate particle physics events much faster than Monte Carlo methods.

In **engineering and robotics**, AI optimization is improving designs of systems ranging from **aerodynamic aircraft shapes** to **microchips**. By coupling simulation with AI (often via *reinforcement learning* or evolutionary algorithms), engineers can automatically discover designs that humans might not conceive. One example is the use of neural networks to optimize rocket engine parts or wind turbine blades for performance and durability. In control systems, AI controllers (like deep reinforcement learning agents) can manage complex tasks – for instance, controlling plasma in fusion reactors (achieved by DeepMind and collaborators) or balancing robots. In the realm of **electrical engineering**, AI helps in designing better semiconductor materials and photonic devices, by predicting how changes at the atomic level affect device performance.

Earth sciences and geoscience benefit from AI in many ways: processing remote sensing data for agriculture or ecology (identifying crop diseases from drone images, mapping wildlife), predicting earthquakes by finding precursory patterns in seismic data, and modeling hydrological systems for water resources management. Each of these involves large, complex datasets where AI can tease out patterns and make forecasts that complement physical models.

Social sciences and other domains: While not always considered "AI4Science" in the narrow sense, AI is increasingly used in economics, sociology, and psychology research to analyze complex social data or simulate scenarios (e.g. agent-based models with learned behaviors). A recent survey even speaks of "AI for social science and social science of AI" [35]. However, these applications often raise different sets of ethical and interpretability questions and are usually regarded as a separate branch (AI for social good, etc.).

To summarize the thematic advances: **AI's core strengths – pattern recognition, prediction, generation, and optimization – have found use in virtually every scientific field.** From unveiling the structure of proteins, to discovering new materials and celestial objects, to optimizing engineering designs and decoding ancient languages (yes, even archaeology and linguistics use AI for decipherment tasks), the landscape of AI4Science is remarkably broad. Interdisciplinary conferences and workshops (NeurIPS now has an "AI for Science" track [36], and ICML 2024 featured an AI4Science workshop) reflect this richness, bringing together diverse research communities. Each community is adapting AI tools to its own needs: sometimes by developing custom AI architectures (like graph networks for molecules, physics-informed nets for fluids), and sometimes by formulating their problems in ways general AI methods can tackle (e.g. treating theorem proving as a game for reinforcement learning).

Notably, a literature analysis by Xie *et al.* (2024) found that the **science and AI research communities often focus on different problems** within AI4Science [37]. Science journals tend to emphasize biomedical and material applications (e.g. protein design, materials discovery), whereas AI conferences explore a broader set of methods and even social or economic domains [37]. Bridging this gap is itself a theme: ensuring that advances in AI reach scientists in all fields, and conversely that AI researchers stay informed of high-impact scientific challenges. Initiatives like cross-disciplinary workshops, joint appointments (AI experts embedded in science departments), and interdisciplinary journals are helping align these efforts.

Role and Transformation of Datasets in AI4Science

Data is the fuel of AI4Science, and how datasets are used or transformed is a pivotal aspect of this field. Scientific data come in many forms – experimental measurements, simulation outputs, images, sequences, scholarly text – and AI4Science often requires innovative ways to leverage these data sources.

A common practice is **dataset repurposing**: using existing scientific datasets for new AI tasks. For instance, decades' worth of simulation data from weather models (the *reanalysis data*) were repurposed to train GraphCast, turning a numerical weather dataset into training fodder for an AI that predicts weather more efficiently [15]. Similarly, the Protein Data Bank's repository of experimentally solved protein structures became the training set for AlphaFold's deep learning model, and vast genomic sequence databases are used to train DNA and protein language models. In many cases, *unlabeled data* is abundant in science (e.g. sequences with unknown function, astronomical images with unknown object labels) – here **self-supervised learning** can extract structure from the data without explicit human annotations [2]. Self-supervised methods (like predicting masked parts of data) have been especially useful in genomics and chemistry, where labeling every data point (such as the function of each gene, or the property of each new molecule) would be infeasible.

Data augmentation is another vital strategy. Scientific datasets can be relatively small or imbalanced (for example, only a few samples of a rare event, or more data for common phenomena than uncommon ones). Researchers therefore generate synthetic data or apply transformations to existing data to expand training sets. In image-based domains (microscopy, radiology), this might mean rotating or adding noise to images.

In chemistry, augmentation could involve generating chemically plausible variants of known molecules to teach a model the relevant chemical space. In simulation-heavy fields, one can run additional simulations under varied conditions as augmented data – effectively infinite data can be produced within computational budget. A notable twist highlighted by Microsoft's fifth paradigm concept is using **simulators as data generators**: since simulation outputs are perfectly labeled by definition, one can simulate thousands of scenarios (limited only by computation) to create a rich dataset, then train an AI emulator on it [23] [24]. This approach turns the problem of sparse real experiments into one of leveraging plentiful synthetic data (with the caveat that the simulator must be trusted to reflect reality).

AI4Science also often involves **integrating multiple datasets or modalities**. Scientific problems rarely hinge on a single type of data – consider drug discovery, where you might have chemical structures, biological assay results, textual literature, and perhaps molecular simulations. Modern AI systems can combine these: for example, a model might take a molecule's graph representation and a protein's sequence or structure and predict a binding affinity. Or an AI agent might read scientific literature (text data) to extract facts that it then uses alongside experimental data to propose new hypotheses [38] [39]. In fact, large language models (LLMs) are now being used to **mine scientific texts and databases**, essentially digesting centuries of recorded knowledge to assist scientists. The U.S. DOE has identified "knowledge distillation" from unstructured data as a key AI capability: an LLM trained on the body of scientific literature could extract every known property of a material or infer cross-disciplinary links that humans haven't noticed [38] [39]. Such a model could then generate new hypotheses – for example, noticing that a certain phenomenon only occurs under specific conditions across papers and suggesting a previously unobserved relationship [40]. This transformation of unstructured data (papers, reports, lab notebooks) into structured knowledge or insights is extremely promising for accelerating discovery.

Benchmark datasets and open data have become an important facet of AI4Science as well. To encourage reproducibility and fair evaluation, researchers are creating standardized datasets for problems like molecule generation (e.g. MOSES, GuacaMol benchmarks), materials discovery, and scientific workflow optimization [41]. An example is the MASSW dataset (2024) which provides a benchmark for AI-assisted scientific workflows [41]. Similarly, the OpenCatalyst dataset released by Facebook AI and CMU – catalysis simulation data comprising hundreds of thousands of examples – has enabled many groups to train models and compare methods on a common ground [20]. The trend is that scientific data, which historically might remain siloed in individual labs, is increasingly shared in machine-readable form to fuel community-wide AI efforts.

When using datasets, researchers must often **transform or encode the data appropriately** for AI models. This can mean converting a molecule's 2D formula into a graph or string format, turning a physical system's state into a vector of features, or normalizing heterogeneous data into a common scale. Domain-specific featurization (like representing molecules with learned embeddings, or encoding periodic boundary conditions for crystal structures) can make a big difference in performance.

Finally, **insight extraction from data** is a goal in itself. Beyond just making predictions, scientists want AI to reveal *why* – e.g. which patterns in genomic data distinguish a disease subtype, or what combination of parameters yields a novel material phase. Unsupervised learning and clustering methods are widely used to let the data "speak." For instance, topological data analysis combined with machine learning was used to find a gene expression "backbone" uniting plant species [42] – an insight that would be hard to obtain without letting an algorithm sift through massive gene datasets for common structure. Such knowledge

discovery aspects of AI4Science are as critical as the predictive aspects: the ultimate aim is not just accurate models, but *new scientific understanding* extracted from data.

In summary, AI4Science practitioners treat data not as static inputs but as something to be **engineered and harnessed creatively**. They merge datasets, simulate new data, convert formats, and apply novel training paradigms (self-supervision, multi-task learning, etc.) to maximize what can be learned. Still, issues of data quality, bias, and access are constant concerns – poor data can mislead even the best model. As the Nature review notes, *"challenges posed by poor data quality and stewardship remain"* across disciplines [3]. Efforts in open-data initiatives, better metadata standards, and interdisciplinary data repositories are ongoing to ensure AI models have reliable fuel for the foreseeable future.

Gaps, Limitations, and Future Directions

Despite remarkable progress, AI4Science is still in its early stages, with many limitations to overcome on the path toward more **general, autonomous scientific AI** (even the aspirational "SuperIntelligent Research Agent" that the question envisions). Key gaps and opportunities include:

- **Bridging the AI–Science Divide:** As discussed, the AI research community and domain science community often operate in silos, prioritizing different problems and methods [37]. Many scientists lack awareness of the latest AI tools, while many AI experts are unfamiliar with high-impact scientific questions [43]. This gap means some powerful AI methods remain underutilized in science, and some scientific challenges remain unsolved due to limited AI attention. Literature surveys found *uneven coverage*: certain scientific problems (e.g. protein folding, materials design) have plenty of AI solutions, whereas areas like **fusion energy, high-energy physics, or geology** might be "under-investigated" by AI [37] [44]. Likewise, some AI techniques (attention mechanisms, advanced RL) haven't been widely tried in scientific contexts [45]. **Implication:** There is a huge opportunity in exploring these under-addressed areas (F1) and discovering new connections between AI methods and scientific problems (F2) [46]. Interdisciplinary collaborations, workshops, and educational programs are needed to cross-pollinate knowledge. The field would benefit from more *"scientist-friendly" AI tools* and more *"science-aware" AI researchers*. Initiatives like Polymathic AI (explicitly aiming to connect disparate fields via a common AI) are one approach [7] [9], and their outcome will be instructive in how to break down domain barriers.

- **Data and Knowledge Challenges:** While data is plentiful in some areas, in others it's scarce or limited in diversity. Many scientific phenomena (especially new or extreme ones) have only small datasets available, which can handicap data-hungry deep learning models. Moreover, scientific data often contains biases – for example, experimental conditions often cluster in certain ranges, or failed/negative results go unreported, skewing the data pool. This raises the need for **AI methods that can learn effectively from limited data**, via techniques like transfer learning (leveraging models trained on related data), few-shot learning, or by incorporating **prior knowledge** to reduce data needs. Indeed, a critical gap in current AI4Science is the integration of *theoretical knowledge and constraints*. Present AI models are largely empirical, learning patterns from data but not necessarily obeying known scientific laws unless explicitly guided. Future AI agents will need to combine the best of data-driven learning with *symbolic reasoning* and *physical principles* – for example, an AI chemistry assistant should respect rules of chemistry (mass balance, valence rules) to avoid suggesting implausible reactions. Early efforts like physics-informed neural networks and hybrid ML-

physics models are steps in this direction, but much more work is needed to ensure AI's outputs are scientifically consistent and credible.

- **Interpretability and Trust:** Scientific users are often understandably cautious about adopting AI conclusions blindly. A major limitation is that many AI models (deep neural nets) are **black boxes**, providing predictions without explanations. In science, understanding *why* a prediction was made can be as important as the prediction itself. This is both a technical and a socio-cultural gap. Technically, more interpretable AI models or post-hoc explanation tools should be developed specifically for scientific data (which might have different notions of explanation than, say, explaining a credit score). For instance, highlighting which part of a molecule led an AI to predict it's toxic, or which features in climate data are signaling an upcoming drought, would greatly increase user trust. Culturally, training scientists to critically evaluate AI outputs and integrate them with domain knowledge is crucial. We need **human-AI collaboration frameworks** where the AI can justify its suggestions in terms scientists find meaningful (perhaps by pointing to analogous known cases or relevant literature, which an LLM-based system could do by citing sources).

- **Generalist AI Scientist vs Narrow Models:** Most current AI4Science systems are still narrow in scope – extremely good at a specific task (e.g. predicting protein structure, or segmenting telescope images) but not able to perform a broader range of scientific reasoning. The "SuperIntelligent Research Agent" would require a **generalist ability**: to observe data, formulate hypotheses, design experiments, draw conclusions, and even adapt its own strategies. We are far from that level of general AI. However, recent projects hint at directions. The **AI Co-Scientist by Google** is one example of a multi-agent system that attempts hypothesis generation in a closed loop [16] [17]. It uses a *"generate, debate, and evolve"* paradigm where different AI agents propose hypotheses, critique each other, and refine ideas, inspired by the scientific method [47]. Initial studies showed it could propose medically relevant hypotheses (like drug repurposing ideas that held up in lab tests) [17]. This suggests that combining large language models (for knowledge and reasoning) with domain-specific models (for simulations/experiments) and a logic for iterative improvement could yield **AI that meaningfully participates in research**. Still, this is a long way from a self-driven Nobel-caliber AI scientist. Limitations include the need for human-provided goals and validation – the AI co-scientist was *"aligned to scientist-provided research objectives and guidance"*, not operating completely autonomously [48]. Truly autonomous science AI would need the ability to *choose its own interesting questions*, which ventures into creativity and curiosity – qualities not easily programmable.

- **Computational and Infrastructural Gaps:** Training state-of-the-art scientific AI models can require enormous computational resources (e.g. GraphCast took four weeks on TPUs to train [15]). Not every scientific group has access to such resources, which could widen a gap between well-funded labs (often Big Tech or elite universities) and others. Democratizing AI4Science means developing more efficient training methods, smaller models that still perform well, or providing better public infrastructure (like high-performance computing centers with AI support). On the infrastructure side, lab automation with AI (robot scientists) requires significant investment in robotics and custom lab integration, which is currently limited to a few labs. Wider adoption will depend on cost reductions and standardization of these systems.

- **Ethical and Quality Considerations:** As AI takes a bigger role in discovery, **ethical issues** arise. For example, if an AI system proposes a new chemical, who is accountable if it turns out to be a dangerous toxin or an environmental hazard? If AI helps generate a new drug, how do we ensure it's

safe – the usual rigorous experimental validation is needed, and AI might help prioritize what to test but should not bypass safety protocols. There's also the risk of *confirmation bias*: if scientists too eagerly believe AI outputs, they might neglect alternative hypotheses (hence the importance of keeping a human-in-the-loop critical mindset). Ensuring **reproducibility** of AI models is another concern – models should be published with enough details or code so that others can reproduce the results on the same data. The scientific community is increasingly expecting this, but it's an ongoing culture change.

On the positive side, these gaps point to **rich future research directions**. One major direction is creating more *general-purpose scientific AI systems* – akin to how GPT-4 is general-purpose for language, we may see "GPT-Sci" that can take in various scientific inputs (text, tables, graphs) and output hypotheses or analyses. Indeed, efforts are underway to train large models on multi-modal scientific data (texts, equations, graphs of experiments) so that the model has a sort of broad scientific literacy. Another direction is enhancing *human-AI synergy*: rather than replace scientists, AI could become a smart assistant that increases a human researcher's capabilities (much like a calculator extends our math ability). To do so, that AI needs to communicate in human-friendly ways (explainable, interactive) and perhaps be personalized to a scientist's field.

The vision of a **SuperIntelligent Research Agent** implies an AI that can *self-evolve*, i.e. improve its own algorithms and knowledge base as it makes discoveries. Some see AutoML and self-play as rudimentary versions of this – where AI designs better AI models (for example, neural architecture search) or improves by competition with itself. In a scientific context, an AI that designs experiments could conceivably identify gaps in its own knowledge and request new experiments to fill them, essentially *learning how to learn* better about the domain. This is analogous to a human scientist noticing their theory doesn't explain something and deciding to investigate that anomaly. Achieving this level of autonomous self-improvement will likely involve advances in meta-learning (AI systems that adapt and learn new problems more efficiently over time) and perhaps integrating reinforcement learning with long-term planning.

In conclusion, the current landscape of AI4Science is vibrant and rapidly advancing, but also **uneven and full of challenges**. There are clear gaps between what AI can do in controlled settings versus what scientists ultimately need in the wild. Yet, the trajectory is toward ever deeper integration. As one perspective article put it, we need "foundational algorithmic approaches that can contribute to scientific understanding or acquire it autonomously" [49] – essentially new AI paradigms that go beyond number-crunching to truly *understand* and *generate* scientific knowledge. The coming years will likely see: more cross-disciplinary alliances (to target neglected problems and share data), improved data curation efforts (because better data is as important as better algorithms), and experimental demonstration of AI systems taking on larger roles in discovery (perhaps an AI autonomously mapping a genome's functionality, or solving a complex physics inverse problem). Each success will move us closer to the vision of an AI that can act as a **genuine partner in science** – or even one day, an independent scientist. For now, human creativity and intuition remain irreplaceable, but with AI as a powerful assistant, the pace of scientific innovation is poised to accelerate in ways we are only beginning to witness [3]. The **opportunity space** – curing diseases faster, discovering sustainable technologies, unraveling cosmic mysteries – makes it imperative to address the current limitations. By doing so, we inch closer to AI systems that not only *support* scientists, but can *themselves drive* the frontier of knowledge forward, marking a fundamentally new era in the scientific endeavor [49].

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